Notes on method of fundamental solutions for solving Laplace equation

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This note is to discuss a method of fundamental solutions (MFS) for solving Laplace equation, based on the recent work by Broms et al.[1]. The system we are interested in is the dielectric spheres excited by a constant electric field.

I. PROBLEM SETUP

The background medium has a dielectric constant $\epsilon_0 = 1$, and the spheres have dielectric constant ϵ_r. Without loss of generality, we assume the electric field is along z-direction, denoted as E_0. The resulting potential $\phi(x)$ satisfies the Laplace equation

$$\begin{cases} \Delta_x \phi(x) = 0 \ , & x \in \Omega_0, \Omega_i \\ \phi^+(x)|_{\partial\Omega_i} = \phi^-(x)|_{\partial\Omega_i} \ , & x \in \partial\Omega_i \\ \epsilon_0 \partial_n \phi^+(x)|_{\partial\Omega_i} = \epsilon_r \partial_n \phi^-(x)|_{\partial\Omega_i} \ , & x \in \partial\Omega_i \\ \phi(x) \to -E_0 z \ , & |x| \to \infty \end{cases} \tag{1}$$

where Ω_0 is the background domain, Ω_i is the domain of the i-th sphere, $\partial\Omega_i$ is the boundary of the i-th sphere, and n is the normal direction pointing from Ω_i to Ω_0.

We then split the potential into two parts, $\phi(x) = u_{inc}(x) + u(x)$, where $u_{inc}(x) = -E_0 z$ is the incident potential, and $u(x)$ is the scattered potential. The scattered potential then satisfies the following equations:

$$\begin{cases} \Delta_x u(x) = 0 \ , & x \in \Omega_0, \Omega_i \\ u^+(x)|_{\partial\Omega_i} = u^-(x)|_{\partial\Omega_i} \ , & x \in \partial\Omega_i \\ \epsilon_0 \partial_n u^+(x)|_{\partial\Omega_i} - \epsilon_0 E_0 n_z = \epsilon_r \partial_n u^-(x)|_{\partial\Omega_i} - \epsilon_r E_0 n_z \ , & x \in \partial\Omega_i \\ u(x) \to 0 \ , & |x| \to \infty \end{cases} \tag{2}$$

where n_z is the z-component of the normal vector.

II. SINGLE SPHERE

We first derive the formula for a single polarizable sphere. For a sphere with radius a centered at the origin of the coordinate system excited by a constant electric field E_0 along z-direction, the potential outside and inside the sphere can be solved analytically as

$$\phi(x) = \begin{cases} -E_0 z + \frac{a^3(\epsilon_r - 1)}{(\epsilon_r + 2)} \frac{E_0 z}{|x|^3} \ , & |x| > a \\ -\frac{3}{\epsilon_r + 2} E_0 z \ , & |x| < a \end{cases} \tag{3}$$

To solve this problem numerically, we make use of the method of fundamental solutions (MFS). For this problem, instead of use $\sigma(x)$ on the surface of the sphere, we place $p(x)$ and $q(x)$ inside and outside the sphere, respectively, which can contribute to the potential outside and inside the sphere, respectively. The idea is similar to the method of images, but the sources now located on sphere with radius $r_p < a$ and $r_q = a^2/r_p > a$. With $p(x)$ and $q(x)$, Eq. 2 can be rewritten as

$$\begin{cases} \int_{\Omega_p} G(x,y) p(y) dy - \int_{\Omega_q} G(x,y) q(y) dy = 0 \ , & |x| = a \\ \int_{\Omega_p} \partial_n G(x,y) p(y) dy - \epsilon_r \int_{\Omega_q} \partial_n G(x,y) q(y) dy = -(\epsilon_r - 1) E_0 n_z \ , & |x| = a \end{cases} \tag{4}$$

Then to discretize the above equations, we use the t-design points on the sphere following Broms et al.[1], which discretize $p(x)$ and $q(x)$ into N equal weights points $\boldsymbol{p}$ and $\boldsymbol{q}$, and also discretize the boundary conditions on M points on the sphere surface, which are denoted as $\boldsymbol{n}_z$. The above equations can be rewritten in matrix form as

$$\begin{cases} \boldsymbol{S}_{pr} \boldsymbol{p} - \boldsymbol{S}_{qr} \boldsymbol{q} = \boldsymbol{0} \\ \boldsymbol{D}_{pr}^T \boldsymbol{p} - \epsilon_r \boldsymbol{D}_{qr}^T \boldsymbol{q} = -(\epsilon_r - 1) E_0 \boldsymbol{n}_z \end{cases} \tag{5}$$

where $\boldsymbol{S}$ and $\boldsymbol{D}^T$ are single and adjoint double layer potential matrices. Eq. (5) can be rewritten as follows:

$$\begin{bmatrix} \boldsymbol{S}_{pr} & \boldsymbol{S}_{qr} \\ \boldsymbol{D}_{pr}^T & \epsilon_r \boldsymbol{D}_{qr}^T \end{bmatrix} \begin{bmatrix} \boldsymbol{p} \\ -\boldsymbol{q} \end{bmatrix} = \begin{bmatrix} \boldsymbol{0} \\ -(\epsilon_r - 1) E_0 \boldsymbol{n}_z \end{bmatrix} \tag{6}$$

which is a $2M \times 2N$ overdetermined linear system, which can be solved by least squares method. Numerical results are shown in Fig. 1, which shows good agreement with the analytical solution.

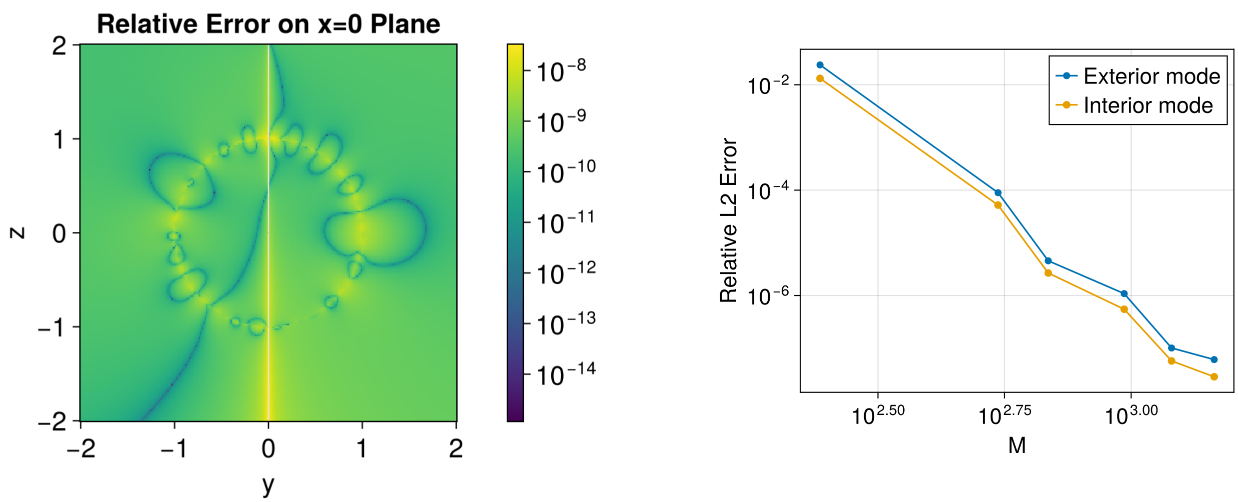

FIG. 1: Numerical results for a single dielectric sphere using MFS.

III. FAR AWAY MULTIPLE SPHERES

We then consider the case of multiple spheres that are far away from each other.

A. Two spheres

For simplicity, we first consider the case of two spheres with the same radius a and dielectric constant ϵ_r, which are centered at $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$, respectively. Again, we assume the electric field is along z-direction, denoted as E_0, thus $u_{inc}(x) = -E_0 z$. The solution is again represented by MFS, where we place $p_i(x)$ and $q_i(x)$ inside and outside the i-th sphere, respectively, as shown in Fig. 2.

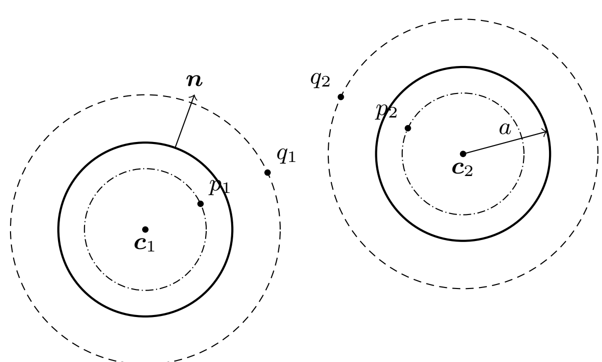

FIG. 2: Two-sphere setup (cross-section in the x–z plane) with MFS source points p_i (inside) and q_i (outside) for each sphere, centers $\boldsymbol{c}_i$, and applied field direction E_0 along $+z$.

The scattered potential can be represented as follows:

$$u(x) = \begin{cases} \int_{\Omega_{p_1}} G(x,y)p_1(y)dy + \int_{\Omega_{p_2}} G(x,y)p_2(y)dy \ , & x \in \Omega_0 \\ \int_{\Omega_{p_2}} G(x,y)p_2(y)dy + \int_{\Omega_{q_1}} G(x,y)q_1(y)dy \ , & x \in \Omega_1 \\ \int_{\Omega_{p_1}} G(x,y)p_1(y)dy + \int_{\Omega_{q_2}} G(x,y)q_2(y)dy \ , & x \in \Omega_2 \end{cases} \tag{7}$$

with which the boundary conditions can be represented as follows:

$$\begin{cases} \int_{\Omega_{p_1}} G(x,y)p_1(y)dy - \int_{\Omega_{q_1}} G(x,y)q_1(y)dy \ = 0, & x \in \partial\Omega_1 \\ \int_{\Omega_{p_2}} G(x,y)p_2(y)dy - \int_{\Omega_{q_2}} G(x,y)q_2(y)dy \ = 0, & x \in \partial\Omega_2 \\ \int_{\Omega_{p_1}} \partial_n G(x,y)p_1(y)dy + (1-\epsilon_r)\int_{\Omega_{p_2}} \partial_n G(x,y)p_2(y)dy - \epsilon_r \int_{\Omega_{q_1}} \partial_n G(x,y)q_1(y)dy = -(\epsilon_r - 1)E_0 n_z \ , & x \in \partial\Omega_1 \\ (1-\epsilon_r)\int_{\Omega_{p_1}} \partial_n G(x,y)p_1(y)dy + \int_{\Omega_{p_2}} \partial_n G(x,y)p_2(y)dy - \epsilon_r \int_{\Omega_{q_2}} \partial_n G(x,y)q_2(y)dy = -(\epsilon_r - 1)E_0 n_z \ , & x \in \partial\Omega_2 \end{cases} \tag{8}$$

Using the same discretization as in the single sphere case, we can rewrite the above equations in matrix form as follows:

$$\begin{bmatrix} \boldsymbol{S}_{p_1r_1} & \boldsymbol{0} & \boldsymbol{S}_{q_1r_1} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{S}_{p_2r_2} & \boldsymbol{0} & \boldsymbol{S}_{q_2r_2} \\ \boldsymbol{D}^T_{p_1r_1} & (1-\epsilon_r)\boldsymbol{D}^T_{p_2r_1} & \epsilon_r \boldsymbol{D}^T_{q_1r_1} & \boldsymbol{0} \\ (1-\epsilon_r)\boldsymbol{D}^T_{p_1r_2} & \boldsymbol{D}^T_{p_2r_2} & \boldsymbol{0} & \epsilon_r \boldsymbol{D}^T_{q_2r_2} \end{bmatrix} \begin{bmatrix} \boldsymbol{p}_1 \\ \boldsymbol{p}_2 \\ -\boldsymbol{q}_1 \\ -\boldsymbol{q}_2 \end{bmatrix} = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{0} \\ -(\epsilon_r - 1)E_0 \boldsymbol{n}_z \\ -(\epsilon_r - 1)E_0 \boldsymbol{n}_z \end{bmatrix} \tag{9}$$

where $\boldsymbol{S}_{ab}$ and $\boldsymbol{D}^T_{ab}$ are the single and adjoint double layer potential matrices for sources at a and targets at b.

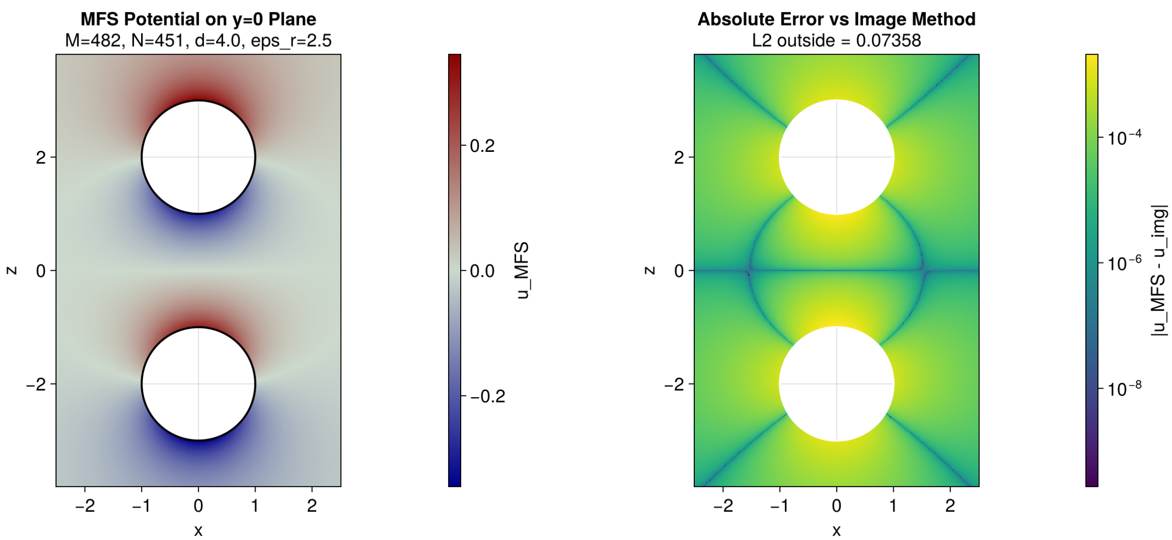

FIG. 3: Potential and error for two spheres aligned along z-axis, the reference result is obtained by an order-1 image approximation.

B. Multiple spheres

We then consider the multiple spheres case. Following the same idea as in the two spheres case, the boundary conditions is given by:

$$\begin{cases} \int_{\Omega_{p_i}} G(x,y)p_i(y)dy - \int_{\Omega_{q_i}} G(x,y)q_i(y)dy \ = 0, & x \in \partial\Omega_i \\ (1-\epsilon_r)\sum_{j\neq i}\int_{\Omega_{p_j}} \partial_n G(x,y)p_j(y)dy + \int_{\Omega_{p_i}} \partial_n G(x,y)p_i(y)dy - \epsilon_r \int_{\Omega_{q_i}} \partial_n G(x,y)q_i(y)dy = -(\epsilon_r - 1)E_0 n_z \ , & x \in \partial\Omega_i \end{cases} . \tag{10}$$

which can be rewritten as the following matrix form after discretization:

$$\begin{bmatrix} \boldsymbol{S}_{p_1r_1} & \boldsymbol{0} & \cdots & \boldsymbol{0} & \boldsymbol{S}_{q_1r_1} & \boldsymbol{0} & \cdots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{S}_{p_2r_2} & \cdots & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{S}_{q_2r_2} & \cdots & \boldsymbol{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{S}_{p_Kr_K} & \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{S}_{q_Kr_K} \\ \boldsymbol{D}^T_{p_1r_1} & (1-\epsilon_r)\boldsymbol{D}^T_{p_2r_1} & \cdots & (1-\epsilon_r)\boldsymbol{D}^T_{p_Kr_1} & \epsilon_r\boldsymbol{D}^T_{q_1r_1} & \boldsymbol{0} & \cdots & \boldsymbol{0} \\ (1-\epsilon_r)\boldsymbol{D}^T_{p_1r_2} & \boldsymbol{D}^T_{p_2r_2} & \cdots & (1-\epsilon_r)\boldsymbol{D}^T_{p_Kr_2} & \boldsymbol{0} & \epsilon_r\boldsymbol{D}^T_{q_2r_2} & \cdots & \boldsymbol{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ (1-\epsilon_r)\boldsymbol{D}^T_{p_1r_K} & (1-\epsilon_r)\boldsymbol{D}^T_{p_2r_K} & \cdots & \boldsymbol{D}^T_{p_Kr_K} & \boldsymbol{0} & \boldsymbol{0} & \cdots & \epsilon_r\boldsymbol{D}^T_{q_Kr_K} \end{bmatrix} \begin{bmatrix} \boldsymbol{p}_1 \\ \boldsymbol{p}_2 \\ \vdots \\ \boldsymbol{p}_K \\ -\boldsymbol{q}_1 \\ -\boldsymbol{q}_2 \\ \vdots \\ -\boldsymbol{q}_K \end{bmatrix} = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{0} \\ \vdots \\ \boldsymbol{0} \\ -(\epsilon_r-1)E_0\boldsymbol{n}_{z,1} \\ -(\epsilon_r-1)E_0\boldsymbol{n}_{z,2} \\ \vdots \\ -(\epsilon_r-1)E_0\boldsymbol{n}_{z,K} \end{bmatrix}, \tag{11}$$

which leads to a $2KM \times 2KN$ overdetermined linear system.

Then we apply the one-body preconditioning technique as in Broms et al.[1]. The self-interaction part of the above matrix is given by

$$\boldsymbol{B} = \begin{bmatrix} \boldsymbol{S}_{pr} & \boldsymbol{S}_{qr} \\ \boldsymbol{D}^T_{pr} & \epsilon_r\boldsymbol{D}^T_{qr} \end{bmatrix}, \tag{12}$$

which is the same for all spheres when they have the same radius and dielectric constant. Reordering the unknown variables as $\boldsymbol{p}_1, -\boldsymbol{q}_1, \boldsymbol{p}_2, -\boldsymbol{q}_2, \cdots, \boldsymbol{p}_K, -\boldsymbol{q}_K$, the above matrix can be rewritten as the following $K \times K$ block system:

$$\begin{bmatrix} \boldsymbol{B}_{11} & \boldsymbol{A}_{12} & \cdots & \boldsymbol{A}_{1K} \\ \boldsymbol{A}_{21} & \boldsymbol{B}_{22} & \cdots & \boldsymbol{A}_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A}_{K1} & \boldsymbol{A}_{K2} & \cdots & \boldsymbol{B}_{KK} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda}_1 \\ \boldsymbol{\lambda}_2 \\ \vdots \\ \boldsymbol{\lambda}_K \end{bmatrix} = \begin{bmatrix} \boldsymbol{b}_1 \\ \boldsymbol{b}_2 \\ \vdots \\ \boldsymbol{b}_K \end{bmatrix}, \tag{13}$$

denoted as $\boldsymbol{G}\boldsymbol{\lambda} = \boldsymbol{b}$, where the reordered unknowns and right-hand sides are defined as follows:

$$\boldsymbol{\lambda}_i = \begin{bmatrix} \boldsymbol{p}_i \\ -\boldsymbol{q}_i \end{bmatrix}, \qquad \boldsymbol{b}_i = \begin{bmatrix} \boldsymbol{0} \\ -(\epsilon_r-1)E_0\boldsymbol{n}_{z,i} \end{bmatrix}, \tag{14}$$

the off-diagonal blocks and diagonal blocks (self-interaction) are

$$\boldsymbol{A}_{ij} = \begin{bmatrix} \boldsymbol{0} & \boldsymbol{0} \\ (1-\epsilon_r)\boldsymbol{D}^T_{p_jr_i} & \boldsymbol{0} \end{bmatrix}, \qquad \boldsymbol{B}_{ii} = \begin{bmatrix} \boldsymbol{S}_{p_ir_i} & \boldsymbol{S}_{q_ir_i} \\ \boldsymbol{D}^T_{p_ir_i} & \epsilon_r\boldsymbol{D}^T_{q_ir_i} \end{bmatrix}. \tag{15}$$

Applying svd to $\boldsymbol{B}$, we have $\boldsymbol{B} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$, and $\boldsymbol{B}^+ = \boldsymbol{V}\boldsymbol{\Sigma}^+\boldsymbol{U}^T$ is the pseudo-inverse of $\boldsymbol{B}$. Then we can precondition the above system as follows:

$$\begin{bmatrix} \boldsymbol{I} & \boldsymbol{A}_{12}\boldsymbol{B}^+ & \cdots & \boldsymbol{A}_{1K}\boldsymbol{B}^+ \\ \boldsymbol{A}_{21}\boldsymbol{B}^+ & \boldsymbol{I} & \cdots & \boldsymbol{A}_{2K}\boldsymbol{B}^+ \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A}_{K1}\boldsymbol{B}^+ & \boldsymbol{A}_{K2}\boldsymbol{B}^+ & \cdots & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \\ \vdots \\ \boldsymbol{\mu}_K \end{bmatrix} = \begin{bmatrix} \boldsymbol{b}_1 \\ \boldsymbol{b}_2 \\ \vdots \\ \boldsymbol{b}_K \end{bmatrix}, \tag{16}$$

denoted as $\hat{\boldsymbol{G}}\boldsymbol{\mu} = \boldsymbol{b}$, where $\boldsymbol{\mu}_i = \boldsymbol{B}\boldsymbol{\lambda}_i$. $\hat{\boldsymbol{G}}$ is now a $2KM \times 2KM$ square matrix, so that the linear system can be solved by iterative methods such as GMRES. Applying the operator $\hat{\boldsymbol{G}}$ to a vector $\boldsymbol{\mu}$ takes the following steps:

1. Compute $\hat{\boldsymbol{\lambda}}_i = \boldsymbol{V}\boldsymbol{\Sigma}^+(\boldsymbol{U}^T\boldsymbol{\mu}_i)$ for each i.
2. Apply the $\boldsymbol{G}$ operator to $\hat{\boldsymbol{\lambda}}$ to get $\boldsymbol{u}_{all} = \boldsymbol{G}\hat{\boldsymbol{\lambda}}$.
3. Substract the diagonal blocks and replace them with identity: $\hat{\boldsymbol{G}}\boldsymbol{\mu} = \boldsymbol{u}_{all} - \boldsymbol{B}_{blkdiag}\hat{\boldsymbol{\lambda}} + \boldsymbol{\mu}$.

For $\boldsymbol{S}_{pr}$, $\boldsymbol{S}_{qr}$ and $\boldsymbol{D}^T_{qr}$, only self-interaction terms are needed, which can be precomputed and applied by direct matrix-vector multiplication. For $\boldsymbol{D}^T_{pr}$, the calculation can be done by one all-to-all FMM with block diagonal corrections.

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[1] A. Broms, A. H. Barnett, and A.-K. Tornberg, Journal of Computational Physics **523**, 113636 (2025).